8. EVALUATING LINE INTEGRAS IN VECTORS

Aim: Using MATLAB Evaluating line integrals

Definition: let $\vec{F}$ be a continuous vector filed defined on a smooth curve C given by a vector function $\vec{r}(t), a \leq t \leq b$. Then the line integral of $\vec{F}$ along C is integral $\int_C \vec{F} \cdot d\vec{r} = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}\,'(t)dt$

inline(expr)	Constructs an inline function object from the MATLAB expression contained in the string expr.
vectorize(fun)	Inserts a . before any ^, * or / in s. The result is a character string
quiver(x,y,u,v)	Displays velocity vectors as arrows with components (u,v) at the points (x,y)
quiver3(x,y,z,u,v,w)	Plots vectors with components (u,v,w) at the points (x,y,z))

Matlab code

```
clc
clear
syms t x y
f=input('Enter the  f vector as i and j order in vector form:');
rbar = input('Enter the r vector as i and j order in vector form:');
lim=input('Enter the limit of integration:');
vecfi=input('Enter the vector field range'); % knowledge of the curve is essential
drbar=diff(rbar,t);
sub = subs(f,[x,y],rbar);
f1=dot(sub,drbar)
int(f1,t,lim(1),lim(2))
P = inline(vectorize(f(1)), 'x', 'y');
Q = inline(vectorize(f(2)), 'x', 'y')
x = linspace(vecfi(1),vecfi(2), 10); y = x;
[X,Y] = meshgrid(x,y);
U = P(X,Y);
V = Q(X,Y);
quiver(X,Y,U,V)
hold on
fplot(rbar(1),rbar(2),[lim(1),lim(2)])
axis on
```

xlabel('x')
ylabel('y')

Example 1

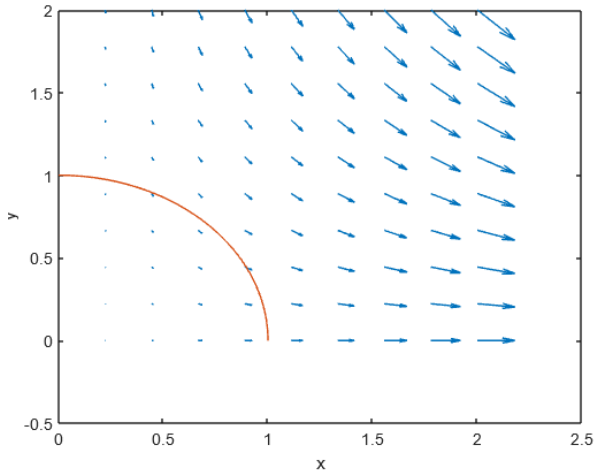
Find the work done by the force field $\vec{F}(x,y) = x^2\vec{\imath} - xy\vec{\jmath}$ in moving a particle along the quarter – circle $\vec{r}(t) = \cos(t)\,\vec{\imath} + \sin(t)\vec{\jmath}$, $0 \leq t \leq \frac{\pi}{2}$

Sol:

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^{\frac{\pi}{2}} \vec{F}(\vec{r}(t)) \cdot \vec{r}\,'(t)dt = \int_0^{\frac{\pi}{2}} (-2cos^2\, t\, \sin t)\, dt$$

In the command window

Enter the f vector as i and j order in vector form:
[x^2 -x*y]
Enter the r vector as i and j order in vector form:
[cos(t) sin(t)]
Enter the limit of integration:
[0 pi/2]
Enter the vector field range
[0 2]
ans = -2/3
In the figure window



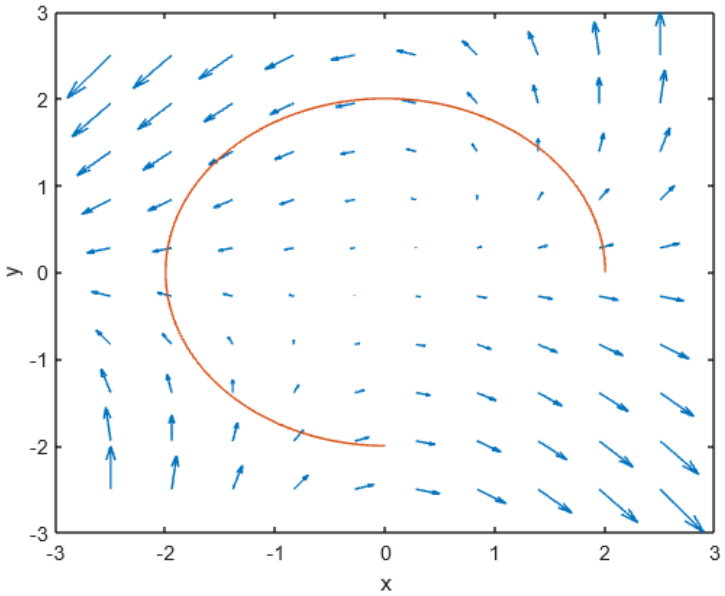
The above figure shows the force field and the curve. The work done is negative because the field impedes movement along the curve.

Example 2

Evaluate the line integral $\int_C \vec{F} \cdot d\vec{r}$, $\vec{F}(x,y) = (x-y)\vec{\imath} + xy\vec{\jmath}$, C is the arc of the circle $x^2 + y^2 = 4$ traversed counter-clockwise from (2, 0) to(0, -2)

In the command window

Enter the f vector as i and j order in vector form:
[x-y x*y]
Enter the r vector as i and j order in vector form:
[2*cos(t) 2*sin(t)]
Enter the limit of integration:
[0, 3*pi/2]
Enter the vector field range
[-2.5 2.5]
ans = 3*pi + 2/3



The above figure shows the force field and the curve. The work done is positive.

Example 3:

Find the work done by the conservative field $\vec{F} = yz\vec{\imath} + xz\vec{\jmath} + xy\vec{k}$ along any smooth curve C joining the point A(-1,3,9) to B(1,6,-4).

Matlab code

```
clc
clear all
syms x y z real
F=input( 'Enter the vector as i,j and k order in vector form:')
pt_A=input( 'Enter the point A(x,y,z) in vector form:');
```

```
pt_B=input( 'Enter the point B(x,y,z) in vector form:');
curl_F = curl(F, [x y z]);
if (curl_F ==[0 0 0]);
 f = potential(F, [x y z])
 f_A=subs(f,[x y z],[pt_A(1) pt_A(2) pt_A(3)]);
 f_B=subs(f,[x y z],[pt_B(1) pt_B(2) pt_B(3)]);
 sol=f_B-f_A
else
 sprintf('curl_F is not equal to zero/ F is not corservative ')
end
```

Command window

Enter the vector as i,j and k order in vector form:[y*z x*z x*y]

Enter the point A(x,y,z) in vector form:[-1 3 9]

Enter the point B(x,y,z) in vector form:[1 6 -4]

f =

x*y*z

sol =

3

Example 4:

Find the work done when a force $\vec{F}(x,y) = (x^2 - y^2 + x)\vec{\imath} - (2xy + y)\vec{\jmath}$ displaces a particle in the xy-plane from(0,0) to (1,1) along the curves (i) $y = x$ (ii) $y^2 = x$. Comment about $\vec{F}$.

Hint: Use **Example 1 Matlab code** for two curves (i) $y = x$ (ii) $y^2 = x$ and comment.

Example 5: Evaluate $\int_C \vec{F} \cdot d\vec{r}$, where $\vec{F}(x,y,z) = (y^2z + 2xz^2)\vec{\imath} + (2xyz)\vec{\jmath} + (xy^2 + 2x^2z)\vec{k}$, and C:$x = \sqrt{t}, y = t + 1, z = t^2, 0 \leq t \leq 1$.

```
clc
clear
syms t x y z
F=input('Enter the  f vector as i and j order in vector form:');
rbar = input('Enter the r vector as i and j order in vector form:');
lim=input('Enter the limit of integration:');
vecfi=input('Enter the vector field range'); % depends on problem
drbar=diff(rbar,t);
sub = subs(F,[x,y,z],rbar);
```

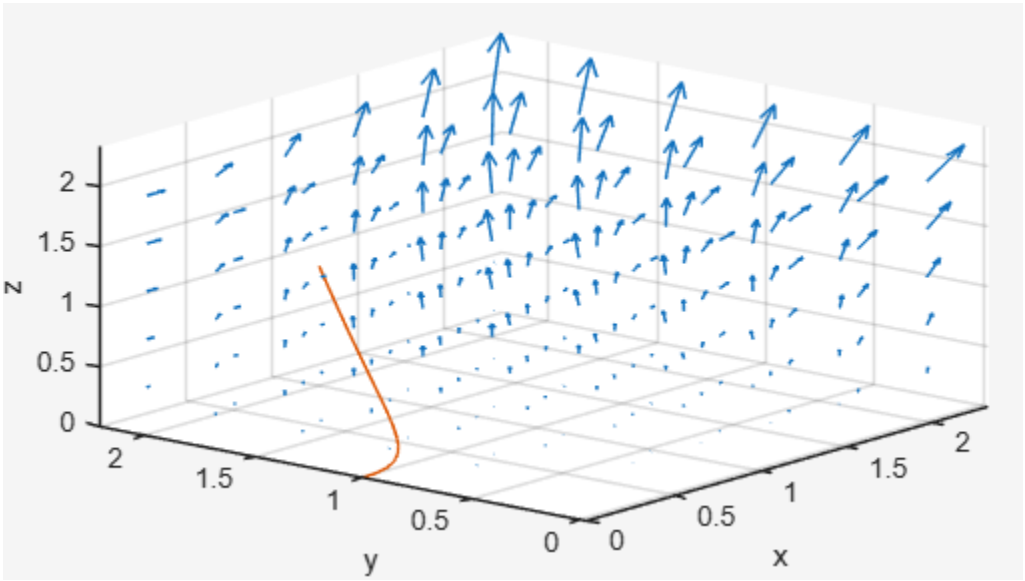
```
f1=dot(sub,drbar)
int(f1,t,lim(1),lim(2))
P = inline(vectorize(F(1)), 'x', 'y','z');
Q = inline(vectorize(F(2)), 'x', 'y','z');
R = inline(vectorize(F(3)), 'x', 'y','z');
x = linspace(vecfi(1),vecfi(2), 7); y = x;
z=x;
[X,Y,Z] = meshgrid(x,y,z);
U = P(X,Y,Z);
V = Q(X,Y,Z);
W = R(X,Y,Z);
quiver3(X,Y,Z,U,V,W,1.5)
hold on
fplot3(rbar(1),rbar(2), rbar(3),[lim(1),lim(2)])
axis on
xlabel('x')
ylabel('y')
zlabel('z')
```

Command window

```
Enter the f vector as i and j order in vector form:[y^2*z+2*x*z^2  2*x*y*z  x*y^2+2*x^2*z]
Enter the r vector as i and j order in vector form:[sqrt(t) t+1  t^2]
Enter the limit of integration:[0 1]
Enter the vector field range[0 2]

ans =

5
```



Exercise

1. Find the work done by the force field $\vec{F}(x,y) = (y+z,\ x+z, x+y)$ on a particle that moves along the line segment (1, 0, 0) to (3, 4, 2)

2. Find the work done by the force field $\vec{F}(x,y) = x\ sin(y)\vec{\imath} + y\vec{\jmath}$ on a particle that moves along the curve $y = x^2$ from (-1, 1) to (2, 4)

3.Evaluate $\int_C \vec{F} \cdot d\vec{r}$, $\vec{F}(x,y,z) = y\ sin(z)\vec{\imath} + z\ sin(x)\vec{\jmath} + x\ sin(y)\vec{k}$ and $\vec{r}(t) = \cos(t)\,\vec{\imath} + \sin(t)\,\vec{\jmath} + \sin(5t)\,\vec{k}$, $0 \leq t \leq \pi$.